

Parallel Transport of Vectors on 3D Mesh by the Vector Heat Method using FEM

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Abstract. Compute the parallel transport of tangent vectors defined on a 3D mesh by the Vector Heat Method using Finite Element Method (FEM).

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1. Introduction

This follows the method described in [3] and [1].

Solve the following Heat Equation PDE with Dirichlet boundary conditions (boundary value problem):

$$\begin{aligned} \text{Solve } \nabla^2 U(\vec{X}, t) &= \frac{\partial U(\vec{X}, t)}{\partial t} && \text{in region } \Omega \\ \text{s.t. } U(\vec{X}, t) &= G(\vec{X}) && \text{on the region boundary } \partial\Omega \text{ for all times} \\ U(\vec{X}, 0) &= U_0(\vec{X}) && \text{initial condition at time } t = 0 \end{aligned}$$

where $U(\vec{X}, t)$ is a vector field and ∇^2 is the “Connection Laplacian” which produces another vector field and is defined in terms of the covariant derivative ∇ as:

$$\nabla^2 U(\vec{X}) = \nabla^* \nabla U(\vec{X}) \tag{1.1}$$

where ∇^* is the adjoint of the covariant derivative ∇ , $U(\vec{X})$ is a tangent vector field on the 3D surface (2D manifold embedded in $\mathbb{R}^3$). $G(\vec{X})$ the vector boundary conditions and $U_0(\vec{X})$ is a vector field at time 0.

The covariant derivative $\nabla(U, w)$ is a “directional derivative” operator that has two vector field arguments: $U(\vec{X})$ and $w(\vec{X})$. Where $U(\vec{X})$ is the vector field which is differentiated and $w(\vec{X})$ is a vector field telling the

direction in which directional derivatives are taken. It returns a vector field as which is a measure of relative change of U w.r.t to w .

Usually one can omit the vector field w in which case we talk about a total derivative ∇U which is a 2-tensor (i.e., a matrix). For example in flat space $\mathbb{R}^3$ the vector field represented as $U = [u_x, u_y, u_z]^T$ the operator ∇U would be:

$$\nabla U = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} u_x & \frac{\partial}{\partial y} u_x & \frac{\partial}{\partial z} u_x \\ \frac{\partial}{\partial x} u_y & \frac{\partial}{\partial y} u_y & \frac{\partial}{\partial z} u_y \\ \frac{\partial}{\partial x} u_z & \frac{\partial}{\partial y} u_z & \frac{\partial}{\partial z} u_z \end{bmatrix}$$

In more compact notation

$$\nabla U = \partial \otimes U = J(U) \quad (1.2)$$

where $\otimes$ is the “tensor product”, $J(U)$ is the Jacobian matrix of U (or the differential) and ∂ is the “gradient operator”.

In a curved manifold the above definition would need to be adjusted to cope with curvature with the aid of a “connection”.

The adjoint of ∇ can be defined by “integration by parts” (see wikipedia) over surface Ω :

Recall adjoint operator is defined as an inner product:

$$\begin{aligned} \langle f, \nabla^* g \rangle &= \langle \nabla f, g \rangle \\ \int_{\Omega} f \nabla^* g \, dx &= \int_{\Omega} \nabla f \, g \, dx \end{aligned}$$

recall intergration by parts formula:

$$\int_{\Omega} f'(x)g(x) \, dx = \int_{\partial\Omega} f(x)g(x) \, dx - \int_{\Omega} f(x)g'(x) \, dx$$

doing integration by parts on $\int \nabla_x f \, g \, dx$ we get:

$$\int_{\Omega} \nabla_X f \, g \, dx = \int_{\partial\Omega} f \, g \, dx - \int_{\Omega} f \, \nabla_X^* g \, dx$$

where ∇_X is the directional derivative such that $\nabla_X f = X \cdot \nabla f$

Applying the above to the vector field valued functions F and G

$$\int_{\Omega} \nabla_X F \cdot G \, dx = \int_{\Omega} F \cdot \nabla_X^* G \, dx + \int_{\partial\Omega} F \cdot G(N) \, dS \quad (1.3)$$

where $\nabla_X F$ and $\nabla_X^* G$ are vector fields and the $(\cdot)$ is inner product. N is unit normal at boundary $\partial\Omega$. For details check A Simple Discretization of the Vector Dirichlet Energy For identities in flat space check adjoints of vector operators

Going back to the original goal it is to compute the parallel transported vector field U from vector fields given at boundary. The Vector Heat Method works as follows: solves the Vector Heat Equation PDE $\frac{\partial U}{\partial t} = \nabla^2 U$ with dirichlet boundary conditions using Finite Element Method (FEM)

$$\begin{aligned}
 &\text{Solve } \Delta U(\vec{X}, t) = \frac{\partial U(\vec{X}, t)}{\partial t} && \text{in region } \Omega \\
 &\text{s.t. } U(\vec{X}, t) = G(\vec{X}) && \text{on the region boundary } \partial\Omega \text{ for all times} \\
 &U(\vec{X}, 0) = U_0(\vec{X}) && \text{initial condition at time } t = 0
 \end{aligned}$$

1.1. The Vector Heat PDE

Since $\frac{\partial U(\vec{X}, t)}{\partial t}$ is a time derivative we discretize the time using Backguard Euler

$$\frac{\partial U(\vec{X}, t)}{\partial t} = \frac{U(\vec{X}, t + \Delta t) - U(\vec{X}, t)}{\Delta t} \quad (1.4)$$

Replacing on original PDE for backguard euler:

$$\frac{U(\vec{X}, t + \Delta t) - U(\vec{X}, t)}{\Delta t} = \nabla^2 U(\vec{X}, t + \Delta t) \quad (1.5)$$

Re-arranging:

$$U(\vec{X}, t + \Delta t) - \Delta t \nabla^2 U(\vec{X}, t + \Delta t) = U(\vec{X}, t) \quad (1.6)$$

or better:

$$U(\vec{X}, t_i) - \Delta t \nabla^2 U(\vec{X}, t_i) = U(\vec{X}, t_{i-1}) \quad (1.7)$$

The integral form (notice $\int_{\Omega}$ is a double integral) is integrating each vector component separately.

$$\int_{\Omega} U(\vec{X}, t_i) dA - \Delta t \int_{\Omega} \nabla^2 U(\vec{X}, t_i) dA = \int_{\Omega} U(\vec{X}, t_{i-1}) dA \quad (1.8)$$

2. Weak Form

The weak form comes from the “variational method”. Here one tries to get a solution to the PDE by translating it into a minimization problem. That’s done by definig an energy function that is a linear functional on a function space Ω . The linear functional is artificial i.e., pure mathematical device to enable optimization. In consequence, solving the variational problem may require less continuity than that of the actual solution (hence the name WEAK form).

Consider the test function $\vartheta(x, y)$ (a.k.a. test function) with compact support defined on Ω . The “Galerkin” variational method states that the inner product of $\vartheta(x, y)$ with a residual function $R(x, y)$ should be zero. Note that in this case ϑ have to be a vector field of the form $\vartheta(\vec{X}) = [\varphi_i(\vec{X}), \varphi_j(\vec{X})]$ which is a vector with a scalar test functions with compact support as components.

$$\int_{\Omega} \vartheta(\vec{X}) \cdot R(\vec{X}) dA = 0$$

where the $(\cdot)$ is inner product and $R(x, y)$ is the residual or error of the function i.e.,

$$R(x, y) = U(\vec{X}, t_i) - \Delta t \nabla^2 U(\vec{X}, t_i) - U(\vec{X}, t_{i-1})$$

The weak form of the PDE is:

$$\begin{aligned} \vartheta(\vec{X}) \cdot R(\vec{X}) &= 0 \\ \vartheta(\vec{X}) \cdot (U(\vec{X}, t_i) - \Delta t \nabla^2 U(\vec{X}, t_i) - U(\vec{X}, t_{i-1})) &= 0 \\ U(\vec{X}, t_i) \cdot \vartheta(\vec{X}) - \Delta t \vartheta(\vec{X}) \cdot \nabla^2 U(\vec{X}, t_i) &= U(\vec{X}, t_{i-1}) \cdot \vartheta(\vec{X}) \end{aligned}$$

and the integral (variational) form of the PDE is

$$\int_{\Omega} U_t \cdot \vartheta \, dA - \Delta t \int_{\Omega} \vartheta \cdot \nabla^2 U_t \, dA = \int_{\Omega} U_{t-1} \cdot \vartheta \, dA \quad (2.1)$$

NOTE on inner products: The integral form is more important since integral of of a “vector field” function $f(x, y)$ times test function $\vartheta(x, y)$ is the “inner product” $\int_{\Omega} f(x, y) \cdot \vartheta(x, y) \, dA$ where the $(\cdot)$ is inner product of vectors. If function $\vartheta(x, y)$ is a “test function” which is a vector field with separate scalar test function with compact support as components then the inner product is way to “measure” the function $f(x, y)$ on location where $\vartheta(x, y)$ is defined.

Replacing $F = \vartheta$ and $G = \nabla U$ in (1.3) we get:

$$\begin{aligned} \int_{\Omega} \nabla \vartheta : \nabla U \, dx &= \int_{\Omega} \vartheta \cdot \nabla^* \nabla U \, dx + \int_{\partial\Omega} \vartheta \cdot \nabla U(N) \, dS \\ \int_{\Omega} \vartheta \cdot \nabla^* \nabla U \, dx &= \int_{\Omega} \nabla \vartheta : \nabla U \, dx - \int_{\partial\Omega} \vartheta \cdot \nabla_N U \, dS \\ \int_{\Omega} \vartheta \cdot \nabla^2 U \, dx &= \int_{\Omega} \nabla \vartheta : \nabla U \, dx - \int_{\partial\Omega} \vartheta \cdot \nabla_N U \, dS \end{aligned} \quad (2.2)$$

where ∇U and $\nabla \vartheta$ are rank-2 tensor fields by virtue of the covariant derivative. Setting $\vartheta = 0$ at the boundary and replacing (2.2) in (2.1)

$$\int_{\Omega} U_t \cdot \vartheta \, dA - \Delta t \int_{\Omega} \nabla \vartheta : \nabla U_t \, dA = \int_{\Omega} U_{t-1} \cdot \vartheta \, dA \quad (2.3)$$

3. Discretization in Flat Space $\mathbb{R}^2$

The domain Ω has to be meshed in N nodes and T triangles. For linear elements the nodes coincides with mesh vertices (but for quadratic elements there will be nodes defined on the triangle’s edge mid-points, so there can be more nodes than vertices).

The “test function” is a function that allows us to linearly interpolate local orthonormal frames located at mesh nodes to the triangle-set surrounding the node. Those local orthonormal frames located at mesh nodes are “tangent spaces” of the nodes. In the flat space of $\mathbb{R}^2$ all tangent spaces are equal, so the interpolation of them is simple. However the key point to understand is that test functions represent linear interpolation (and atenuation) of “local

frames". We define a test function φ_i per node N_i which will be used to linearly interpolate tangent vectors defined in terms of local frames located at nodes to all triangles incident to the node N_i .

Since we are in flat space $\mathbb{R}^2$, the tangent space at node N_i is the same as any other tangent space everywhere else in the domain Ω . Therefore parallel transport is just the identify function or "teleportation", and the covariant derivative $\nabla_X U$ coincides with the directional derivative $X \cdot \nabla U$.

The "local basis" for tangent space at N_v is defined as $T_v = (e_1, e_2)$, which is an orthonormal frame. The "local basis" for tangent space at triangle is $T_f = (e_1, e_2)$, the connection 1-form $\omega_i^f = I$ where I is the identity matrix. Therefore the vectors at $U(x, y)$ are expressed in terms of the "global basis" (e_1, e_2)

As usual, we define a set of scalar tests functions with compact support $\phi_i(x, y)$ for every node N_i to be equal to 0 or 1 and linear everywhere else:

$$\begin{aligned}\phi_j(\vec{N}_j) &= 1 \quad \forall j = i \\ \phi_j(\vec{N}_j) &= 0 \quad \forall j \neq i \\ \phi_j(\vec{X}) &= \text{linear in } \Omega_i\end{aligned}$$

We use those ϕ_j to define the vector-valued basis function φ_i (see ref1, ref2, ref3, ref4):

$$\begin{aligned}\varphi_i(\vec{X}) &= \vec{\phi}_i(\vec{X}) T_j \\ \varphi_i(\vec{X}) &= \begin{bmatrix} \phi_i(\vec{X}) \\ \phi_i(\vec{X}) \end{bmatrix}^T (e_1, e_2)\end{aligned}\tag{3.1}$$

Where $T_j = (e_1, e_2)$ is the "local basis" (2×2 matrix) for tangent space at N_j , $\vec{X}$ is vector expressed also in basis (e_1, e_2) .

Now we discretise the solution $U(x, y)$ to be a finite sum:

$$U(x, y) = \sum_j^N \varphi_j(x, y)^T \begin{bmatrix} U_j^{e_1} & 0 \\ 0 & U_j^{e_2} \end{bmatrix}\tag{3.2}$$

where $\odot$ is the component-wise product of vectors (a.k.a., Hadamard product) and U_j is a vector (2×1 column).

Now we discretise $\nabla U(x, y)$ which is a rank-2 tensor field

$$\begin{aligned}\nabla U(x, y) &= \sum_j^N \nabla \varphi_j(x, y) U_j \\ \nabla U(x, y) &= \sum_j^N \begin{bmatrix} \nabla \phi_j(\vec{X}) \\ \nabla \phi_j(\vec{X}) \end{bmatrix}^T \begin{bmatrix} U_j^{e_1} & 0 \\ 0 & U_j^{e_2} \end{bmatrix}\end{aligned}\tag{3.3}$$

where $\nabla \varphi_j$ is a 2×2 matrix.

Given $\vartheta(x, y) = [\phi_i(x, y), \phi_i(x, y)]^T$, we discretise the test function $\nabla\vartheta(x, y)$ in the following manner:

$$\begin{aligned}\nabla\vartheta(x, y) &= \sum_j^N \nabla\psi_j(x, y) \\ \nabla\vartheta(x, y) &= \sum_j^N \begin{bmatrix} \nabla\phi_j(\vec{X}) \\ \nabla\phi_j(\vec{X}) \end{bmatrix}\end{aligned}\quad (3.4)$$

We discretise the $\nabla\vartheta : \nabla U$:

$$\begin{aligned}\nabla\vartheta : \nabla U &= \left(\sum_i \nabla\psi_i \right) : \left(\sum_j \nabla\varphi_j U_j \right) \quad (3.5) \\ \nabla\vartheta : \nabla U &= \sum_{ij} (\nabla\psi_i : \nabla\varphi_j) U_j \\ \nabla\vartheta : \nabla U &= \sum_{ij} \text{Tr} \left(\begin{bmatrix} \nabla_{e_1}\phi_i(\vec{X}) & \nabla_{e_2}\phi_i(\vec{X}) \\ \nabla_{e_1}\phi_i(\vec{X}) & \nabla_{e_2}\phi_i(\vec{X}) \end{bmatrix}^T \begin{bmatrix} \nabla_{e_1}\phi_j(\vec{X}) & \nabla_{e_2}\phi_j(\vec{X}) \\ \nabla_{e_1}\phi_j(\vec{X}) & \nabla_{e_2}\phi_j(\vec{X}) \end{bmatrix} U_j \right) \\ \nabla\vartheta : \nabla U &= \sum_k \sum_{ij} \left(\nabla\phi_i(\vec{X}) \cdot \nabla\phi_j(\vec{X}) \right) U_j^{e_k}\end{aligned}$$

where the contraction of the two rank-2 tensors is another rank-2 tensor e.g., $G_i \cdot G_j = G_i^T G_j$ or in tensor notation $G_{ij} G_{jk} = G_{ik}$. The integrals are now much smaller i.e., per triangle-set Ω_i around nodes $\vec{N}_i$ or tent function $\varphi_i(\vec{X})$

The stiffness matrix for triangle T is:

$$M_k(i, j) = I_{2 \times 2} \int_{\Omega} \nabla\phi_i \cdot \nabla\phi_j \, dA \quad (3.6)$$

So we can form the linear system:

$$\sum_{ij} I_{2 \times 2} \left(\int_{\Omega} \nabla\phi_i \cdot \nabla\phi_j \, dA \right) U_j \quad (3.7)$$

where U_j is 2×1 column vector and $I_{2 \times 2}$ is identity matrix.

4. Assembly

The process of creating the stiffness matrix M is called assembly. It consist of creating the matrix M_k for each triangle T_k and then add it to the global M matrix.

The global matrix M is $2N \times 2N$ where N is number of mesh nodes. In previous section 3 the matrix M_k is a $2N \times 2N$ a super-sparse block matrix where each entry is 2×2 block matrix just with a few non-zero entries which corresponds to a single triangle contribution.

$$M = \sum_k M_k \quad (4.1)$$

In practice the M_k matrix is not $2N \times 2N$ super-sparse, but a 6×6 matrix (in case of linear triangle elements). There is a map from indices of the 6×6 matrix to the $2N \times 2N$ matrix as follows.

Each triangle's nodes has a global index assigned in the mesh, we also establish a local indexing of triangle like 1, 2 and 3. We define a trivial bijective map between global and local indices.

5. Discretization in 3D curved space e.g., 3D surface

The 3D curved domain Ω has to be discretised in N nodes and T triangles. For linear elements the nodes coincides with vertices, but for quadratic elements there will nodes defined on the mid-point of edges of triangles, so there will be more nodes than vertices.

We define a test function Φ_i per node N_i which will be used to linearly interpolate basis tangent vectors located at mesh nodes to all triangles incident to the node N_i .

First we define a local basis for tangent space T_v per node N_i in which tangent vector of $U(x, y)$ will reside. We will also need to define a local tangent space basis per each triangle T_f , which not coincide with T_v , that will be used to interpolate basis tangent vectors within triangles. Finally we need a "connection" R_i^f which parallel transports a basis of tangent vectors T_v at node N_i to a basis of tangent vectors T_f at triangle T_f incident to node N_i :

$$R_i^f = \begin{array}{ccc} T_f^T & Q_{vf} & T_v \\ 2 \times 2 & 2 \times 3 & 3 \times 3 \end{array} \quad 3 \times 2 \quad (5.1)$$

where:

R_i^f is 2×2 matrix connecting 2 tangent spaces: the local tangent space at node N_i to a local tangent space at triangle T_f . The tangent spaces are connected in such a way that a tangent vector is "parallel transported" from one tangent space to the next

T_f is the 3×2 local tangent space basis for triangle f : $T_f = [e_i, e_j]^T$

Q_{vf} is the 3×3 rotation matrix aligning node N_i tangent plane with face T_f plane.

T_v is the 3×2 local tangent space basis for tangent space at node N_i : $T_v = [e_i, e_j]^T$

The connection R_i^f defines the mapping Φ_i^f at node N_i restricted to triangle f :

$$\Phi_i^f(\vec{X}) = R_i^f \vec{X} \quad (5.2)$$

which parallel transport a tangent vector $\vec{X}$ in tangent space basis at node N_i to a triangle T_f local basis.

We also define scalar “tests” functions with compact support $\phi_i(x, y)$ for every node N_i to be equal to 0 or 1 and linear everywhere else:

$$\begin{aligned}\phi_j(\vec{N}_j) &= 1 \quad \forall j = i \\ \phi_j(\vec{N}_j) &= 0 \quad \forall j \neq i \\ \phi_j(\vec{X}) &= \text{linear in } \Omega_i\end{aligned}$$

We use those ϕ_j to define the vector-valued basis function Ψ_i^f :

$$\begin{aligned}\Psi_i^f(x, y)(\vec{X}) &= \phi_j(x, y)\Phi_i^f \vec{X} \\ \Psi_i^f(x, y)(\vec{X}) &= \phi_j(x, y)R_i^f \vec{X}\end{aligned}\tag{5.3}$$

Now we discretise the solution $U(x, y)$ on triangle T_f to be a finite sum:

$$\begin{aligned}U(x, y) &= \sum_j \Psi_j(x, y)U_j \\ U(x, y) &= \sum_j \phi_j(x, y)R_j^f U_j\end{aligned}\tag{5.4}$$

and the covariant derivative $\nabla U(x, y)$:

$$\begin{aligned}\nabla U(x, y) &= \sum_j \nabla \Psi_j(x, y)U_j \\ \nabla U(x, y) &= \sum_j \nabla \phi_j(x, y) \otimes R_j^f U_j\end{aligned}\tag{5.5}$$

or in matrix form:

$$\nabla U(x, y) = \sum_j R_j^f \begin{bmatrix} U_j^{e_1} & 0 \\ 0 & U_j^{e_2} \end{bmatrix} \begin{bmatrix} \nabla \phi_j(x, y) \\ \nabla \phi_j(x, y) \end{bmatrix}\tag{5.6}$$

which is a 2×2 matrix.

We discretise the test function $\vartheta(x, y)$ on triangle T_f as:

$$\begin{aligned}\vartheta(x, y) &= \sum_j R_j^f \psi_j(x, y) \\ \vartheta(x, y) &= \sum_j R_j^f [\phi_j(x, y), \phi_j(x, y)]^T\end{aligned}\tag{5.7}$$

where $\psi_j(x, y) = [\phi_j(x, y), \phi_j(x, y)]^T$ and its covariant derivative $\nabla \vartheta(x, y)$ as:

$$\begin{aligned}\nabla \vartheta(x, y) &= \sum_j R_j^f \nabla \psi_j(x, y) \\ \nabla \vartheta(x, y) &= \sum_j R_j^f [\nabla \phi_j(x, y), \nabla \phi_j(x, y)]^T\end{aligned}\tag{5.8}$$

We discretise the $\nabla \vartheta : \nabla U$ for each element T_f on the triangle-set around node N_i as contribution of nodes N_j . Using the fact that $A : B =$

$Tr(A^T B)$ and the cyclic property of the trace we get:

$$\begin{aligned}
 \nabla \vartheta : \nabla U &= \left(\sum_i R_i \nabla \psi_i \right) : \left(\sum_j \nabla \phi_j \otimes R_j U_j \right) \\
 \nabla \vartheta : \nabla U &= \sum_{ij} R_i \nabla \psi_i : \nabla \phi_j \otimes R_j U_j \\
 \nabla \vartheta : \nabla U &= \sum_{ij} Tr(\nabla \psi_i^T R_i^T R_j [U_j] \nabla \psi_j) \\
 \nabla \vartheta : \nabla U &= \sum_{ij} Tr(\nabla \psi_j \nabla \psi_i^T R_i^T R_j [U_j]) \quad (5.9)
 \end{aligned}$$

Where $[U_j] = \begin{bmatrix} U_j^{e_1} & 0 \\ 0 & U_j^{e_2} \end{bmatrix}$. Noting that $Tr(\nabla \psi_j \nabla \psi_i^T) = Tr(\nabla \psi_i^T \nabla \psi_j) = \nabla \phi_i \cdot \nabla \phi_j$:

$$\nabla \vartheta : \nabla U = \sum_k \sum_{ij} (\nabla \phi_i \cdot \nabla \phi_j) (R_i^T R_j U_j) \cdot e_k \quad (5.10)$$

The stiffness matrix for triangle T is a 6×6 matrix:

$$M_k(i, j) = R_i^T R_j \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j \, dA \quad (5.11)$$

So we can form the linear system:

$$\sum_{ij} \left(\int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j \, dA \right) R_i^T R_j U_j \quad (5.12)$$

where U_j is 2×1 column vector and $R_i^T R_j$ is the discrete connection 2×2 matrix.

6. The Covariant Derivative $\nabla \Psi$

We use the Cartan's moving frames formalism. Moving frames are orthonormal frames attached to each point on the surface. Those frames lies in the tangent space associated to each point, thus the frames are orthogonal unit tangent vectors. Those frames don't form a "holonomic" basis i.e., those are not vector fields associated to a directional derivative or partial derivative operator. In particular, moving frames are not continuous over the surface and its length is always 1 (in contrast, holonomic basis can have different lengths, thus their inner product defines the metric tensor). For this reason the Christoffel Symbols is a concept associated to differentiation of holonomic basis while "Connection Forms" is a concept associated with "exterior differentiation" of moving frames. However in the flat space $\mathbb{R}^n$ under cartesian coordinates the Christoffel Symbols and Connection Forms are the same.

First recap definition of covariant derivative of a vector field U in direction of vector field w (see the Levi-Civita connection):

$E = (e_1, e_2)$	local frame field i.e., moving frame seen as pair of tangent vectors e_1 and e_2
$U = U_1 e_1 + U_2 e_2$	vector field expressed in moving frames as $\sum_i U_i e_i$
$w = w_1 e_1 + w_2 e_2$	vector field expressed in moving frames as $\sum_i w_i e_i$

$$\omega(e_k) = \begin{bmatrix} \omega_{11}(e_k) & \omega_{12}(e_k) \\ \omega_{21}(e_k) & \omega_{22}(e_k) \end{bmatrix} \text{ Matrix of 1-forms a.k.a. connection form}$$

We denote ∂U_i the gradient of scalar component U_i which is a vector $\partial U_i = \left[\frac{\partial}{\partial e_1} U_i, \frac{\partial}{\partial e_2} U_i \right]^T$. And ∂U the Jacobian matrix of U

$$\begin{aligned} \partial U &= [\partial U_1, \partial U_2]^T \\ \partial U &= \begin{bmatrix} \frac{\partial}{\partial e_1} U_1 & \frac{\partial}{\partial e_2} U_1 \\ \frac{\partial}{\partial e_1} U_2 & \frac{\partial}{\partial e_2} U_2 \end{bmatrix} \end{aligned} \quad (6.1)$$

The covariant derivative $\nabla_w U$ of a vector field U in direction of vector field w is defined as

$$\begin{aligned} \nabla_w U &= w \cdot \nabla U \\ \nabla_w U &= w_1 \nabla_{e_1} U + w_2 \nabla_{e_2} U \end{aligned} \quad (6.2)$$

Now we need to expand $\nabla_{e_i} U$:

$$\begin{aligned} \nabla_{e_j} U &= \partial U e_j + \omega(e_j) U \\ \nabla_{e_j} U &= \begin{bmatrix} \partial U_1 \cdot e_j + \omega_{11}(e_j) U_1 + \omega_{12}(e_j) U_2 \\ \partial U_2 \cdot e_j + \omega_{21}(e_j) U_1 + \omega_{22}(e_j) U_2 \end{bmatrix} \end{aligned} \quad (6.3)$$

Putting back $w = w_1 e_1 + w_2 e_2$ we get:

$$\nabla_w U = \partial U w + \omega(w) U \quad (6.4)$$

The above expression coincides with definition in Liu, Tong, De Goes, Desbrun “Discrete Connection and Covariant Derivative for Vector Field Analysis and Design” where $\partial U w$ is the multiplication of the Jacobian of U with w which is giving the directional derivative in flat space. $\omega(w) U$ is the multiplication of the connection form $\omega(w)$ with U which is parallel transporting U to the tip of w .

We can define ∇U just in terms of U without a particular direction w as argument (see covariant derivative of a vector field).

$$\begin{aligned} \nabla_w U &= \partial_{e_k} U_i w_k + \omega_{ij}^k w_k U_i \\ \nabla_w U &= (\partial_{e_k} U_i + \omega_{ij}^k U_i) w_k \\ \nabla U &= \partial_{e_k} U_i + \omega_{ij}^k U_i \end{aligned} \quad (6.5)$$

which can also be written as:

$$\nabla U = \partial U + \omega_{ij}^k U_i \quad (6.6)$$

With the above representation we can contract ∇U with vector w to get the classic Levi-Civita connection:

$$\nabla_w U = w \cdot \nabla U \quad (6.7)$$

Recall the definition of $\Psi(x, y)$ basis function which is defined for every point (x, y) on triangle T_f applied to a tangent vector $\vec{X}_i$ which is defined at node N_i :

$$\Psi_i(x, y)(\vec{X}_i) = \phi_i(x, y)R_i^f \vec{X}_i \quad (6.8)$$

The basis function Ψ does a parallel transport of vectors from tangent space at N_i to a flat tangent space at triangle T_f by using a discrete connection R_i^f . The discrete connection R_i^f is in essence the Connection Form $\omega(\vec{X}_i)$. The covariant derivative $\nabla\Psi$ is simply:

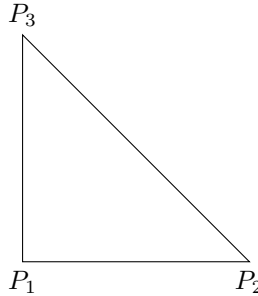
$$\begin{aligned} \nabla\Psi_i(x, y)(\vec{X}_i) &= \nabla\phi_i(x, y) \otimes R_i^f \vec{X}_i + \phi_i(x, y)\nabla(R_i^f \vec{X}_i) \\ \nabla\Psi_i(x, y)(\vec{X}_i) &= \partial\phi_i(x, y) \otimes R_i^f \vec{X}_i \end{aligned} \quad (6.9)$$

where $\partial\phi_i(x, y)$ is the gradient of a scalar valued function $\phi_i(x, y)$, which is a vector and $\otimes$ is the tensor (or dyadic) product of vectors. $\nabla(R_i^f \vec{X}_i) = \partial(R_i^f \vec{X}_i) = 0$ as $R_i^f \vec{X}_i$ is constant vector field inside triangle tangent space (triangle is flat $\mathbb{R}^2$ space).

7. Reference Elements

One of the novelties of FEM is to be able to compute the integrals per individual elements on an easy and generic way by using so called Reference Elements.

For triangle elements we just need a single reference triangle



$$P_1 = (0, 0)$$

$$P_2 = (1, 0)$$

$$P_3 = (0, 1)$$

The P1 (linear) 2D Lagrange shape functions $\psi(u, v)$ has the form

$$\psi(u, v) = Au + Bu + C$$

For the reference triangle defined above the $\psi(u, v)$ takes the simple form:

$$\begin{aligned}\psi_1(u, v) &= 1 - u - v \\ \psi_2(u, v) &= u \\ \psi_3(u, v) &= v\end{aligned}$$

We can check that $\psi_i \cdot \psi_j = \delta_{ij}$

$$\begin{aligned}\psi_1(P_1) &= 1 & \psi_1(P_2) &= 0 & \psi_1(P_3) &= 0 \\ \psi_2(P_1) &= 0 & \psi_2(P_2) &= 1 & \psi_2(P_3) &= 0 \\ \psi_3(P_1) &= 0 & \psi_3(P_2) &= 0 & \psi_3(P_3) &= 1\end{aligned}$$

We define a function $F(\vec{U})$, a.k.a., *pushforward*, that maps reference 2D triangle to "physical" 2D triangle on the mesh (the inverse $F^{-1}(\vec{X})$, a.k.a., *pullback*, maps triangle from physical to reference frame).

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Which can be expressed in vector form

$$\begin{aligned}F(\vec{U}) &= B\vec{U} + \vec{X}_1 \\ F^{-1}(\vec{X}) &= B^{-1}(\vec{X} - \vec{X}_1)\end{aligned}$$

We denote ψ_i^k the function ψ_i defined on physical triangle K .

$$\begin{aligned}\psi_i^k(\vec{X}) &= \psi_i \circ F^{-1}(\vec{X}) \\ \psi_i^k(\vec{X}) &= \psi_i(F^{-1}(\vec{X}))\end{aligned}$$

The gradient of $\psi_i^k(\vec{X})$ in physical triangle is:

$$\begin{aligned}\nabla_{\vec{X}} \psi_i^k(\vec{X}) &= \nabla_{\vec{X}} \psi_i(F^{-1}(\vec{X})) \\ &= \nabla_{\vec{U}} \psi_i(\vec{U}) \cdot \nabla_{\vec{X}} F^{-1}(\vec{X}) \\ &= B^{-T} \nabla_{\vec{U}} \psi_i(\vec{U})\end{aligned}$$

where we used $\vec{U} = F^{-1}(\vec{X})$ and $\nabla_{\vec{X}} F^{-1}(\vec{X}) = B^{-T}$.

The gradient of $\psi_i(\vec{U})$ in reference triangle is:

$$\begin{aligned}\nabla \psi_1 &= (-1, -1) \\ \nabla \psi_2 &= (1, 0) \\ \nabla \psi_3 &= (0, 1)\end{aligned}$$

So the gradient of $\psi_i^k(\vec{X})$ in physical triangle k is "constant":

$$\begin{aligned}\nabla\psi_1^k &= B^{-T}(-1, -1) \\ \nabla\psi_2^k &= B^{-T}(1, 0) \\ \nabla\psi_3^k &= B^{-T}(0, 1)\end{aligned}$$

7.1. For 3D Triangles

For 3D triangles the map function F changes as follows:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \\ z_2 - z_1 & z_3 - z_1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} + \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$$

In matrix algebra it can be expressed as:

$$\begin{aligned}F(\vec{U}) &= B\vec{U} + \vec{X}_1 \\ F^{-1}(\vec{X}) &= (B^T B)^{-1} B^T (\vec{X} - \vec{X}_1)\end{aligned}$$

or

$$F^{-1}(\vec{X}) = B^{-*}(\vec{X} - \vec{X}_1)$$

where $B^{-*} = (B^T B)^{-1} B^T$ is the Moore-Penrose pseudo-inverse of B .

The same analysis as before holds so the gradient in “physical” triangle k is:

$$\begin{aligned}\nabla\psi_1^k &= B^{-*T}(-1, -1) \\ \nabla\psi_2^k &= B^{-*T}(1, 0) \\ \nabla\psi_3^k &= B^{-*T}(0, 1)\end{aligned}$$

8. Numerical Quadrature

The entries of the matrix M_T^k are the following double integrals:

$$M_T^k(i, j) = \int_{T^k} \nabla\psi_i^k(\vec{X}) \cdot \nabla\psi_j^k(\vec{X}) dA$$

indices (i, j) run for all nodes of triangle T .

Let define the inner product of basis functions as

$$I_{ij}^k(\vec{X}) = \nabla\psi_i^k(\vec{X}) \cdot \nabla\psi_j^k(\vec{X})$$

The 3-point Gauss quadrature rule for the linear triangle is:

$$M_T^k(i, j) = \frac{area(T^k)}{3} (I_{ij}^k(P_1) + I_{ij}^k(P_2) + I_{ij}^k(P_3)) \quad (8.1)$$

evaluated at the vertices of the triangle P_1 , P_2 and P_3 . Which is an exact quadrature formula for linear function on a triangle. However since $\nabla\psi_i^k(\vec{X})$ does not depends on $\vec{X}$ the quadrature expression reduces to:

$$M_T^k(i, j) = \text{area}(T^k) \nabla\psi_i^k(\vec{X}) \cdot \nabla\psi_j^k(\vec{X})$$

9. Implementation

Implementation can be found in [2]

References

- [1] H. P. Langtangen. Introduction to finite element methods. 2014.
- [2] M. C. Lopez Belon. Project on github. Available at <https://github.com/mauricioccele/basic-fem-heat-2d>, 2024.
- [3] F.-J. Sayas. A gentle introduction to the finite element method. 2015.

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